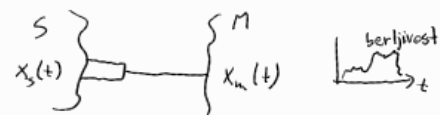


Optimalno filtriranje

- predpis za optimalno sinhronizacijo med S in M



zahteve:

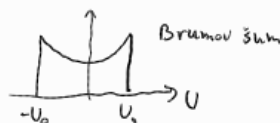
- šibek sklopilov M-S
- berljivost $X_m(t)$ v M
- $\langle (x_s - x_m)^2 \rangle$
- dinamika x_s zib bolj podobna dinamiki x_m

Brunov žum

$$U = U_0 \cos \omega t$$

$$\frac{dP}{dt} = \text{konst} = \frac{1}{T/2}$$

$$\frac{dP}{dU} = \frac{dP}{dt} \left(\frac{dU}{dt} \right)^{-1} = \frac{2}{T} \cdot \frac{1}{\omega U_0 \sqrt{1 - \cos^2 \omega t}} = \frac{2}{T} \cdot \frac{1}{2\pi} \cdot \frac{1}{\sqrt{U_0^2 - U^2}} = \frac{1}{\pi} \cdot \frac{1}{\sqrt{U_0^2 - U^2}}$$



Merilni žum

$$z = x + r \quad \frac{dP}{dr} = \frac{1}{\sqrt{2\pi}} \frac{1}{\sigma} e^{-r^2/2\sigma^2} \quad \langle r \rangle = \int_{-\infty}^{\infty} \frac{dP}{dr} r dr = 0$$

• multiplika. sprejemljiva
• porazd. po $N(0, \sigma)$

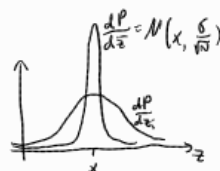
$$\text{Var}(r) = \langle (r - \langle r \rangle)^2 \rangle = \langle r^2 \rangle = \int_{-\infty}^{\infty} \frac{dP}{dr} r^2 dr = \sigma^2$$

$$u = \frac{r}{\sigma} \sim N(0, 1)$$

standardizirano
normalna porazd.

Povprečevanje

$$\bar{z} = \frac{1}{N} \sum z_i, \quad \frac{dP}{dz_i} = N(x, 0)$$



$$\frac{dP}{dz} = ?$$

$$\langle z_i - x \rangle = 0 = \langle r \rangle$$

$$\langle \bar{z} - x \rangle = 0$$

$$\langle (\bar{z} - x)^2 \rangle = \langle \left(\frac{1}{N} \sum (z_i - x) \right)^2 \rangle = \frac{1}{N^2} \left(\sum \langle (z_i - x)^2 \rangle + \underbrace{\sum \sum_{i \neq j} \langle (z_i - x)(z_j - x) \rangle}_{\langle r_i r_j \rangle = 0} \right) = \frac{1}{N^2} \sum \langle r_i^2 \rangle = \frac{\sigma^2}{N}$$

Zdravstvenje meritev

$$z_1, \sigma_1; z_2, \sigma_2 \rightarrow x, \sigma = ?$$

$$\text{celina: } \frac{(z_1 - x)^2}{\sigma_1^2} + \frac{(z_2 - x)^2}{\sigma_2^2} = 2J(x) = \min.$$

$$\rightarrow 0 = \frac{d}{dx} (2J(x)) = -\frac{2(z_1 - x)}{\sigma_1^2} - \frac{2(z_2 - x)}{\sigma_2^2}$$

$$x \left(\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2} \right) = \frac{z_1}{\sigma_1^2} + \frac{z_2}{\sigma_2^2} \rightarrow x = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2} z_1 + \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} z_2$$

$$\sigma_1 < \sigma_2:$$

$$\bar{z} = z_1 + \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} (z_2 - z_1)$$

naša stara ocena z_1 + $\underbrace{\frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} (z_2 - z_1)}_{\text{inovacija}}$

Ali je to res optimalno?

$$z_1 = x + r_1$$

$$z_2 = x + r_2$$

$$z = \alpha z_1 + \beta z_2 = x + r_3 \quad \alpha + \beta = 1, \quad r_3 = \alpha r_1 + \beta r_2$$

$$= \alpha(x + r_1) + \beta(x + r_2) =$$

$$= \underbrace{(\alpha + \beta)x}_{=x} + \alpha r_1 + \beta r_2$$

$$\langle r_3^2 \rangle = \langle (\alpha r_1 + (1-\alpha)r_2)^2 \rangle =$$

$$= \alpha^2 \langle r_1^2 \rangle + (1-\alpha)^2 \langle r_2^2 \rangle + 2\alpha(1-\alpha) \langle r_1 r_2 \rangle =$$

$$= \alpha^2 \sigma_1^2 + (1-\alpha)^2 \sigma_2^2 \quad 0, \text{ ker nekorrelirani}$$

$$\rightarrow \frac{d}{d\alpha} \langle r_3^2 \rangle = 2\alpha \sigma_1^2 + 2(1-\alpha)(-\sigma_2^2) = 0$$

$$\alpha = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}$$

$$\beta = 1 - \alpha = \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2}$$

Korelacija in kovarianca

- 2 seta meritev x_i, y_i
- Čns. povezava med 3m. 1. in 2. meritve $\Rightarrow$ meritvi sta korelirani

$$u_x = x - \bar{x} \quad \bar{u}_x = 0 \quad \overline{u_x^2} = \sigma_x^2$$
$$u_y = y - \bar{y} \quad \bar{u}_y = 0 \quad \overline{u_y^2} = \sigma_y^2$$

Kovarianca $\sigma_{xy} = \overline{u_x u_y} = \overline{u_x u_y}$, $\boxed{\rho_{xy} \leq 1}$ korelacija

$$\sigma_{xy} = \frac{1}{N} \sum (x_i - \bar{x})(y_i - \bar{y}) = \frac{1}{N} \sum (x_i y_i - \bar{x} y_i - \bar{y} x_i + \bar{x} \bar{y}) = \frac{1}{N} (\sum x_i y_i - \bar{x} \sum y_i - \bar{y} \sum x_i + N \bar{x} \bar{y}) =$$
$$= \frac{1}{N} \sum x_i y_i - \bar{x} \bar{y} - \bar{y} \bar{x} + \bar{x} \bar{y} = \overline{x y} - \bar{x} \bar{y}$$

$$\sigma_{x2} = \rho \sigma_1 \sigma_2 = \overline{w_1 w_2}$$
$$w_1 = \alpha w_2 + w, \quad \overline{w w_2} = 0$$

$$z_1 - x = \alpha(z_2 - x) + w, \quad \overline{w(z_2 - x)} = 0$$

$$\rho \sigma_1 \sigma_2 = \overline{(\alpha w_2 + w) w_2} = \alpha \overline{w_2^2} + \overline{w w_2} = \alpha \sigma_2^2$$
$$\rightarrow \alpha = \rho \frac{\sigma_1}{\sigma_2}$$
$$\sigma_1^2 = \overline{w_1^2} = \overline{(\alpha w_2 + w)^2} = \alpha^2 \overline{w_2^2} + \overline{w^2} + 2\alpha \overline{w w_2} = \alpha^2 \sigma_2^2 + \sigma_w^2$$
$$\sigma_w^2 = \sigma_1^2 - \rho^2 \frac{\sigma_1^2}{\sigma_2^2} = \sigma_1^2 (1 - \rho^2)$$

Optimalno združena ocena:

$$2J(x) = \frac{w_1^2}{\sigma_1^2} + \frac{w_2^2}{\sigma_2^2} = \frac{(w_1 - \alpha w_2)^2}{\sigma_w^2} = \dots = \frac{(z_2 - x)^2}{(1 - \rho^2) \sigma_2^2} + \frac{(z_1 - x)^2}{(1 - \rho^2) \sigma_1^2} - \frac{2\rho(z_1 z_2 - x(z_1 + z_2) + x^2)}{(1 - \rho^2) \sigma_1 \sigma_2}$$

$$\frac{d}{dx} (2J(x)) = \frac{-2(z_2 - x)}{(1 - \rho^2) \sigma_2^2} - \frac{2(z_1 - x)}{(1 - \rho^2) \sigma_1^2} - \frac{2\rho(-z_1 + z_2 + 2x)}{(1 - \rho^2) \sigma_1 \sigma_2} = 0$$
$$x \left[\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2} - \frac{2\rho}{\sigma_1 \sigma_2} \right] = \frac{z_2}{\sigma_2^2} + \frac{z_1}{\sigma_1^2} - \frac{\rho(z_1 + z_2)}{\sigma_1 \sigma_2} \rightarrow x = \dots$$

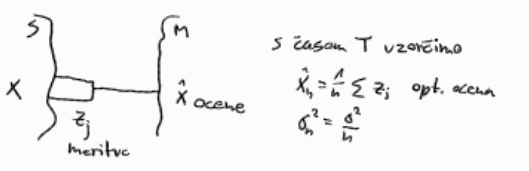
x_{opt} za korelirane količine

Če $\rho = 0$, pridemo do istega rezultata kot za nekorelirane količine V

$$\rho = 1: x = \left(\frac{\sigma_1^2 + \sigma_2^2 - 2\sigma_1 \sigma_2}{\sigma_1^2 \sigma_2^2} \right)^{-1} \left[\frac{z_1 \sigma_2^2 + z_2 \sigma_1^2 - (z_1 + z_2) \sigma_1 \sigma_2}{\sigma_1^2 \sigma_2^2} \right] = (\sigma_1 - \sigma_2)^{-2} (\sigma_1 / \sigma_2) (z_1 \sigma_2 - z_2 \sigma_1) \stackrel{z_2 = z_1}{=} z_1 = z_2$$

$$\sigma_1 = \sigma_2: x = \left(\frac{2 - 2\rho}{\sigma^2} \right)^{-1} \left[\frac{z_1 + z_2 - \rho(z_1 + z_2)}{\sigma^2} \right] = \frac{1}{2} (z_1 + z_2)$$

Sprotno sledenje konst. količin



$$z_j = x + v_j$$

Šum nekoreliran $\rightarrow \langle v_i v_j \rangle = \delta_{ij} \sigma^2$

$$\langle v_i v_j \rangle = ? \quad \sigma_{n+1}^2 = \sigma_n^2 + \sigma^2$$

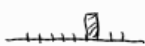
$$t = nT \quad \hat{x}_{n+1} = \frac{1}{n+1} \left(\sum_{i=1}^n z_i + z_{n+1} \right) = \frac{n}{n+1} \hat{x}_n + \frac{1}{n+1} z_{n+1} = \frac{n+1-1}{n+1} \hat{x}_n + \frac{1}{n+1} z_{n+1} =$$
$$\downarrow$$
$$(n+1)T \quad = \hat{x}_n + \frac{1}{n+1} (z_{n+1} - \hat{x}_n)$$
$$\rightarrow \hat{x}_{n+1} = \hat{x}_n + \frac{\sigma_{n+1}^2}{\sigma^2} (z_{n+1} - \hat{x}_n)$$

Prehod v zvezno sliko:

$$\lim_{T \rightarrow 0} \frac{\hat{x}_{n+1} - \hat{x}_n}{T} = \dot{\hat{x}}(t) = \frac{\sigma_{n+1}^2}{\sigma^2 T} (z_{n+1} - \hat{x}_n)$$
$$\lim_{T \rightarrow 0} \frac{\hat{x}_n}{T} = \hat{x}(t)$$
$$\sigma_n^2 \rightarrow \sigma_x^2(t)$$
$$z_n \rightarrow z(t)$$
$$v_n \rightarrow v(t)$$
$$\langle v(t) v(t') \rangle = \delta(t - t') R \quad \text{nekoreliranost}$$

za $t \ll T$ je manjški šum koreliran
za $t \gg T$ je manjški šum nekoreliran
 $\ll T$ kar. čas fluktuacije:

$$\langle r_i r_j \rangle = \delta_{ij} \sigma^2$$



$$\sum \langle r_i r_j \rangle = \dots = \sigma^2 T$$

$$\langle r(t) r(t') \rangle = \delta(t-t') R$$



$$\int \langle r(t) r(t') \rangle dt = \dots = R$$

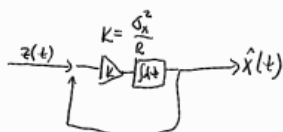
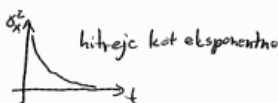
$$R = \lim_{T \rightarrow 0} (\sigma^2 T)$$

$$\rightarrow \dot{\hat{x}}(t) = \frac{\sigma_x^2}{R} (z - \hat{x})$$

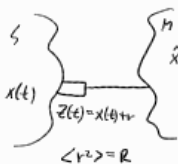
Disperzija v zvezi sliki

$$\frac{\sigma_{n+1}^2 - \sigma_n^2}{T} = \frac{1}{T} \left(\frac{\sigma_n^2}{\sigma_n^2 + \sigma^2} - \frac{\sigma_n^2 (\sigma_n^2 + \sigma^2)}{\sigma_n^2 + \sigma^2} \right)$$

$$\dot{\sigma}_x^2(t) = - \frac{(\sigma_x^2(t))^2}{\sigma_n^2 T \sigma^2} = - \frac{\sigma_x^2}{R}$$



Merjenje skalarne spremljivke



ne poznamo dinamike $x(t)$

$$\rightarrow \hat{x}(t) = z(t)$$

$$\hat{\sigma}_x^2(t) = R$$

poznamo dinamiko:

$$\dot{x}(t) = Ax + c(t) \quad \text{lin. dif. en. 1. reda}$$

$$\langle v(t) v(t') \rangle = R \delta(t-t')$$

$$\rightarrow \text{diskretno: } x_n = x(nT)$$

$$x_{n+1} = \phi_n + (A(nT)x_n + c(nT))T =$$

$$= \underbrace{x_n (1 + A(nT)T)}_{\phi_n} + \underbrace{c(nT)T}_{c_n}$$

• enako dinamiko privzamemo tudi za $\hat{x}$

• lin. dinamika ohranja obliko Gauss. porazila

$$t = nT: \hat{x}_n, \hat{\sigma}_n^2$$

$$t = (n+1)T: \text{Napoved ocene: } \bar{x}_{n+1} = \phi_n \hat{x}_n + c_n \quad \text{Napoved ocene}$$

$$\bar{\sigma}_{n+1}^2 = \phi_n^2 \hat{\sigma}_n^2$$

$$\text{• dokaz: } \bar{\sigma}_{n+1}^2 = \langle (\bar{x}_{n+1} - x_{n+1})^2 \rangle = \langle (\phi_n \hat{x}_n + c_n - (\phi_n x_n + c_n))^2 \rangle =$$

$$= \langle \phi_n^2 (\hat{x}_n - x_n)^2 \rangle = \phi_n^2 \hat{\sigma}_n^2$$

• izostranec ocena za $(n+1)T$:

$$\hat{x}_{n+1} = \bar{x}_{n+1} + \frac{\hat{\sigma}_{n+1}^2}{\sigma^2} (z_{n+1} - \bar{x}_{n+1}) \quad \leftarrow \text{ostrežnje z meritvijo}$$

$$\hat{\sigma}_{n+1}^2 = \bar{\sigma}_{n+1}^2 + \sigma^2$$

Novc oznake

$$\hat{\sigma}_{nn}^2 \rightarrow P_{n+1} \quad \text{kovarianca iz stare ocene}$$

$$\bar{\sigma}_{nn}^2 \rightarrow M_{n+1} \quad \text{kovarianca napovedi}$$

$$\frac{\hat{\sigma}_{nn}^2}{\sigma^2} \rightarrow K_{n+1} \quad \text{ojačevalni faktor}$$

Dinamičen šum

$$x_{n+1} = \phi_n x_n + c_n + \underbrace{g_n w_n}_{\text{multiplikativni faktor}}$$

$$\langle w \rangle = 0 \quad \text{dinamični šum}$$

$$\langle w_i w_j \rangle = \underbrace{Q \delta_{ij}}_{\text{vektorski šum}}$$

• naključni nekorrelirani šumi - v sistemu n : $\bar{x}_{n+1} = \phi_n \hat{x}_n + c_n$

$$M_{n+1} = \langle (\bar{x}_{n+1} - x_{n+1})^2 \rangle =$$

$$= \langle (\phi_n (\hat{x}_n - x_n) - g_n w_n)^2 \rangle = \langle \phi_n^2 (\hat{x}_n - x_n)^2 + g_n^2 w_n^2 - 2 \phi_n g_n w_n (\hat{x}_n - x_n) \rangle$$

$$M_{n+1} = \phi_n^2 P_n + g_n^2 Q$$

Prehod v zvezno sliko

$$\hat{x}_{n+1} = \bar{x}_{n+1} + K_{n+1}(\bar{z}_{n+1} - \bar{x}_{n+1})$$

$$P_{n+1} = P_{nn} - \frac{P_{nn}^2}{P_{nn} + \sigma^2} = \phi_n^2 P_n + \Gamma_n^2 Q_n - \frac{(\phi_n^2 P_n + \Gamma_n^2 Q_n)^2}{P_{nn} + \sigma^2}$$

$$\frac{\hat{x}_{n+1} - \hat{x}_n}{T} = \frac{\phi_n \hat{x}_n + c_n - x_n + K_{n+1}(\bar{z}_{n+1} - \bar{x}_{n+1})}{T} =$$

$$= \underbrace{\frac{(\phi_n - 1)}{T} \hat{x}_n}_{A(t)} + \underbrace{\frac{c_n}{T}}_{\hat{x}(t)} + \underbrace{\frac{P_{n+1}}{\sigma^2 T} (\bar{z}(t) - \hat{x}(t))}_{C(t)}$$

$$\dot{\hat{x}}(t) = A(t) \hat{x}(t) + C(t) + \frac{P(t)}{R} (\bar{z}(t) - \hat{x}(t))$$

$$\dot{P} = \frac{P_{nn} - P_n}{T} = 2AP(t) + \Gamma^2 Q - \frac{P^2}{R}$$

Vektorska spremljivka

$$\begin{cases} \hat{x} = x + p_x \\ \hat{v} = v + p_v \\ \bar{x} = x + m_x \\ \bar{v} = v + m_v \end{cases} \quad \begin{matrix} \dot{x} = v \\ \dot{v} = \Sigma F/m \end{matrix}$$

$$\rightarrow \frac{d}{dt} \begin{bmatrix} x \\ v \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix} + \dot{C}$$

Linearno osterenje

$$\hat{x} = \bar{x} a_{xx} + \bar{v} a_{xv} + b_x \bar{z} = a_{xx}(x + m_x) + a_{xv}(v + m_v) + b_x(x + r) \rightarrow x + p_x = x \underbrace{(a_{xx} + b_x)}_1 + \underbrace{a_{xv} v + a_{xx} m_x + b_x r}_0 = x + \underbrace{a_{xx} m_x + b_x r}_{p_x}$$

$$\langle p_x^2 \rangle = a_{xx}^2 \langle m_x^2 \rangle + (1 - a_{xx})^2 \langle r^2 \rangle + 2 a_{xx} \langle m_x r \rangle =$$

$$= a_{xx}^2 \sigma_x^2 + (1 - a_{xx})^2 \sigma_r^2$$

$$\frac{d}{da_{xx}} \langle p_x^2 \rangle = 2 a_{xx} \sigma_x^2 - 2(1 - a_{xx}) \sigma_r^2 = 0$$

$$\rightarrow a_{xx} = \frac{\sigma_r^2}{\sigma_x^2 + \sigma_r^2}$$

$$v + p_v = \hat{v} = a_{vx}(x + m_x) + a_{vv}(v + m_v) + b_v(x + r) = \underbrace{(a_{vx} + b_v)}_0 x + a_{vv} m_v + v + m_v + b_v r$$

$$p_v = a_{vx} m_x + m_v + b_v r = a_{vx} (m_x + r) + m_v$$

$$\langle p_v^2 \rangle = \dots = a_{vx}^2 [\langle m_x^2 \rangle + \langle r^2 \rangle] + \langle m_v^2 \rangle + 2 a_{vx} \langle m_x m_v \rangle$$

$$\frac{d}{da_{vx}} \langle p_v^2 \rangle = 0 = 2 a_{vx} (\sigma_x^2 + \sigma_r^2) + 2 \langle m_x m_v \rangle$$

$$a_{vx} = - \frac{\langle m_x m_v \rangle}{\sigma_x^2 + \sigma_r^2}$$

$$\rightarrow \hat{x} = a_{xx} \bar{x} + (1 - a_{xx}) \bar{z} = \dots = \bar{x} + \frac{\sigma_x^2}{\sigma_x^2 + \sigma_r^2} (\bar{z} - \bar{x})$$

$$\hat{v} = a_{vx} \bar{x} + a_{vv} \bar{v} - a_{vx} \bar{z} = \bar{v} + \frac{\langle m_x m_v \rangle}{\sigma_x^2 + \sigma_r^2} (\bar{z} - \bar{x})$$

Čeprav merimo samo lego, ostrino dve količini

Osterenje vekt. meritev

$$\begin{cases} \hat{x} = x \\ \bar{x} = x + m \\ \bar{z} = Hx + r \end{cases} \quad \begin{matrix} \langle m m^T \rangle = M \text{ kov. matrika} \\ H = M^T \\ \langle r r^T \rangle = R \text{ matrika šuma} \end{matrix}$$

$$x + \hat{p} = \hat{x} = A \bar{x} + B \bar{z}$$

matrika
u teži

$$\langle \hat{p} \hat{p}^T \rangle = P \text{ kov. matrika izostreane ocene}$$

$$x + \hat{p} = A \bar{x} + B \bar{z} = A(x + m) + B(Hx + r) =$$

$$= \underbrace{(A + BH)}_I x + A m + B r$$

$$I + A + BH = I \text{ vez}$$

$$P = \langle (A m + B r)(A m + B r)^T \rangle = \dots = A M A^T + B R B^T$$

Optimizacija P

- minimiziramo trP

$$\tilde{P} = A M A^T + B R B^T - (A + BH - I) \lambda$$

$$\frac{\partial \tilde{P}}{\partial A} \rightarrow \frac{\partial \tilde{P}}{\partial A} = 0$$

$$\rightarrow A M + A M - X^T = 0$$

$$2 A M - X^T = 0$$

$$\frac{\partial \tilde{P}}{\partial B} = 0$$

$$2 B R - (H \lambda)^T = 0$$

$$\frac{\partial \tilde{P}}{\partial \lambda} = 0$$

$$(A + BH - I)^T = 0$$

$$\rightarrow A = \frac{\lambda}{2} M^{-1}$$

$$A^T = M^{-1} \frac{\lambda}{2}$$

$$B = \frac{(H \lambda)^T}{2} R^{-1}$$

$$B^T = R^{-1} H \frac{\lambda}{2}$$

$$A^T + H^T B^T = I$$

$$M^{-1} \frac{\lambda}{2} + H^T R^{-1} H \frac{\lambda}{2} = I$$

$$(M^{-1} + H^T R^{-1} H) \frac{\lambda}{2} = I$$

$$\frac{\lambda}{2} = (M^{-1} + H^T R^{-1} H)^{-1}$$

$$\rightarrow P = \frac{\lambda^T}{2} M^{-1} M M^{-1} \frac{\lambda}{2} + \frac{\lambda^T}{2} H^T R^{-1} R R^{-1} H \frac{\lambda}{2} =$$

$$= \frac{\lambda^T}{2} \underbrace{(M^{-1} + H^T R^{-1} H)}_{\frac{\lambda^T}{2}} \frac{\lambda}{2} = \frac{\lambda^T}{2} = \frac{\lambda}{2}$$

$$\rightarrow \begin{cases} P^{-1} = M^{-1} + H^T R^{-1} H \\ P = M - M H^T (R + H M H^T)^{-1} H M \end{cases}$$

Izostreva ocena

$$\hat{x} = A\bar{x} + Bz = \frac{\lambda^T}{2} M^{-1} \bar{x} + \left(H \frac{\lambda}{2} \right)^T R^{-1} z =$$

$$= \frac{\lambda^T}{2} M^{-1} \bar{x} + \frac{\lambda^T}{2} H^T R^{-1} H \bar{x} - \frac{\lambda^T}{2} H^T R^{-1} H \bar{x} + \frac{\lambda^T}{2} H^T R^{-1} z =$$

$$= \frac{\lambda^T}{2} \left[\underbrace{(M^{-1} + H^T R^{-1} H)}_{\left(\frac{\lambda^T}{2} \right)^{-1}} \bar{x} + H^T R^{-1} (z - H \bar{x}) \right] =$$

$$\hat{x} = \bar{x} + P H^T R^{-1} (z - H \bar{x})$$

Prehod v zv. sliko

$$H: \hat{x}_t \rightarrow \bar{x}_{t+1} = \phi_t \hat{x}_t + c_t$$

$$S: x_{t+1} = \phi_t x_t + c_t + \Gamma_t w_t$$

$$\begin{aligned} M_{t+1} &= \phi_t P_t \phi_t^T + \Gamma_t Q_t \Gamma_t^T \quad Q_t = \langle w_t w_t^T \rangle \\ P_{t+1} &= P_{t+1} - P_{t+1} H^T (H P_{t+1} H^T + R_{t+1})^{-1} H P_{t+1} \\ \bar{x}_{t+1} &= \phi_t \hat{x}_t + c_t \\ \hat{x}_{t+1} &= \bar{x}_{t+1} + P_{t+1} H^T R_{t+1}^{-1} (z_{t+1} - H \bar{x}_{t+1}) \end{aligned}$$

$$\frac{\hat{x}_{t+1} - \hat{x}_t}{T} = \dot{\hat{x}}(t) = \frac{\bar{x}_{t+1} - P_{t+1} H^T R_{t+1}^{-1} (z_{t+1} - H \bar{x}_{t+1}) - \hat{x}_t}{T} =$$

$$= \underbrace{\frac{\phi_{t+1} - 1}{T}}_{A(t)} \hat{x}_t + \underbrace{\frac{c_t}{T}}_{c(t)} - \underbrace{P(t) H^T \frac{R_{t+1}^{-1}}{T}}_{R(t)^{-1}} (z(t) - H \hat{x}(t))$$

$$\dot{\hat{x}} = A \hat{x} + c(t) - P H^T R^{-1} (z - H \hat{x})$$

$$P_{t+1} - P_t = P_{t+1} - P_{t+1} H^T (H P_{t+1} H^T + R_{t+1})^{-1} H P_{t+1} - P_t =$$

$$= \underbrace{\phi_t P_t \phi_t^T + \Gamma_t Q_t \Gamma_t^T}_{(A+AT)}$$

$$= P_t^T A P_t + P_t A^T + A P_t A^T - P_t^T =$$

$$\rightarrow \dot{P} = A P + P A^T + \underbrace{\frac{\Gamma_t Q_t \Gamma_t^T}{T}}_{\Gamma(t) Q(t) \Gamma^T(t)} - \underbrace{\frac{P_{t+1} H^T (H P_{t+1} H^T + R_{t+1})^{-1} H P_{t+1}}{T}}_{P H^T R(t)^{-1} H P}$$

$$\rightarrow \dot{P} = A P + P A^T + \Gamma Q \Gamma^T - P H^T R^{-1} H P$$